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MCB 203 — Past Questions & Materials

by Renance

Course: MCB 203

Institution: Osun State University, Osogbo

Papers in this pack: 1

Levels: 200

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Paper 1 of 1 — MCB 203 PQ&ANS

Level 200 • First Semester • PastQuestion

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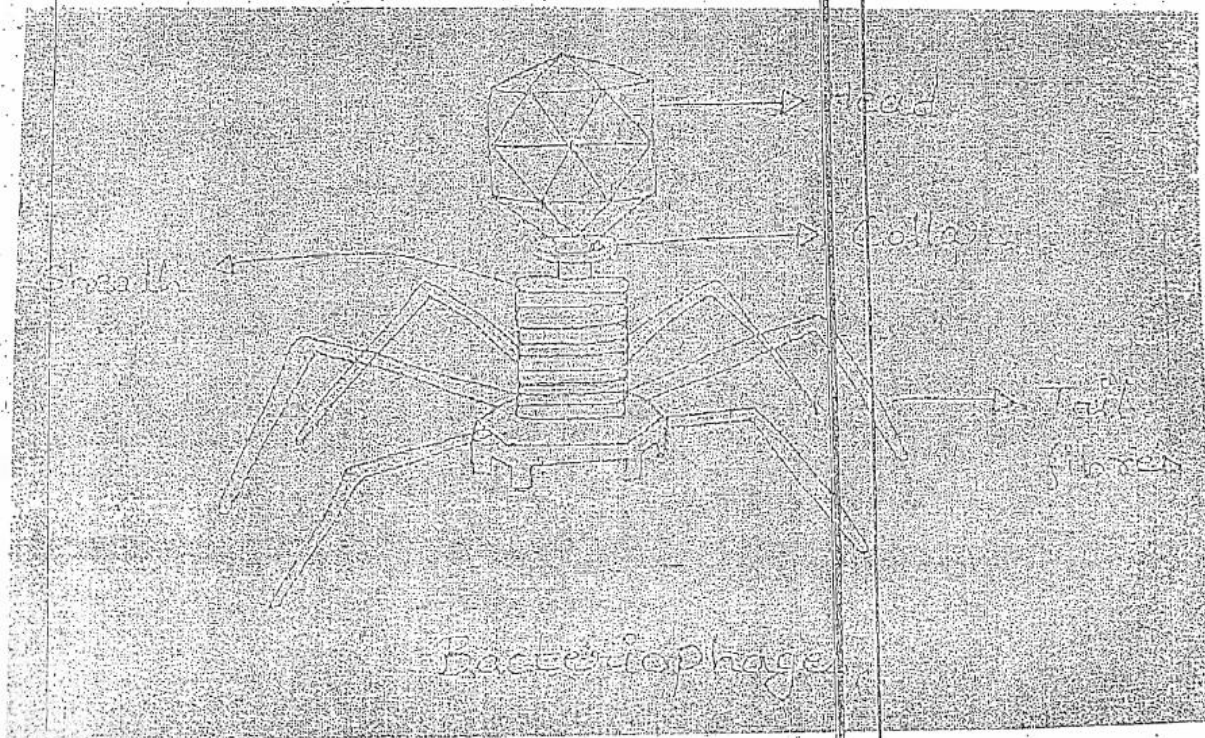
MCB 203 (VIROLOGY)

2017/2018

- 1 a. Draw and label a typical phage
- b. State the roles played by any four of the parts shown in question 1a in the phages replication cycle
- c. Draw any three morphologies of named plants virions.
- d. Mention 3 scientist and state their contribution to the field of Virology.

Answer.

1a:



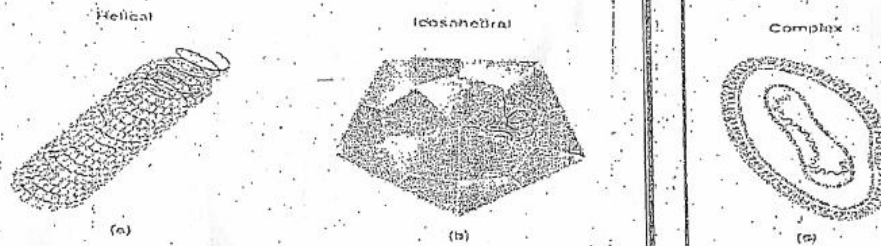
b. Tail fibres - they are used to attach to the cell wall of the host.

Head - to protect the genome of the phage

Sheath - it helps to create a pathway for viral genome delivery.

Collar - help to attach six long tail fibers to the phage particle during the virus assembly and control the retractions of the long tail fibers under conditions unfavorable for infection.

c. Helical



di. 1796, Edward Jenner used cowpox lesion to vaccinate against small pox. He was the first to deliberately vaccinate against any infectious disease.

ii. Louis Pasteur (1855) experimented the term with rabies Vaccination and originated the Term virus (Latin name for poison) and Vaccination in honour of Edward Jenner.

iii. Frederick Leoffler and Paul Frosch demonstrated FMD (Foot and Mouth disease) which was cured by TMV as filterable agent.

e. State four differences between bacteria and Viruses:

Bacteria	Viruses
Bacteria are single-celled, living organisms	Viruses are not considered to be "living" because they require a host cell to survive long-term, for energy, and to reproduce.

Less than 1 % of bacteria cause disease. Most are beneficial for our good health and the health of Earth's ecosystems.	Most viruses causes disease.
They can be inhibited or killed by antibiotics	They can't be killed by antibiotics.
They have a cell wall and all the components necessary to survive and reproduce, although some may derive energy from other sources.	Viruses consist of only one piece of genetic material and a protein shell called a capsid. They survive and reproduce by "hijacking" a host cell, and using its ribosomes to make new viral proteins.

2a Briefly describe transcription, translation, genome replication and assembly of Double stranded DNA phage in its bacteria host.

Answer

*Transcription; Transcription of a D.S DNA phage in its bacteria host would ignite DNA dependent RNA polymerase which would transcribe the phage DNA into the corresponding mRNA

*Translation; The ribosome of the bacteria cell would translate the mRNA of the phage into different proteins of the phage which folds into Capsid, Collar, E.T.C

* Genome replication; Genome of the phage possess so many genes (all sort of chromosome) also the genetic material of the protein possess gene which will also possess DNA dependant RNA polymerase which will then replicate the genome of the phage (Viral enzyme)

* Assembly; like attract like, assembling of the component part occurs

b. Mention any 5 plant Viruses and their Cognate hosts

Answer

Ebola virus – Human

Human immunodeficiency virus – Human

Tobacco mosaic virus – Tobacco

Potato virus X – Potato

Cucumber mosaic virus – Cucumber

c. State any five (5) insects and the plant Viruses of each vectors.

Insects	Plant virus
Aphids	Potyvirus
Leafhoppers	Niloparvata lugens (NLRV)
Whiteflies	Begomoviruses
Planthoppers	Southern rice black-streaked dwarf virus (SRBSDV)
Thrips	Tospovirus

d. Why do Virologist generally refers to Viruses as "Filterable agent"?

Answer.

The term "filterable viruses" was introduced in the late nineties of the preceding century, to denote a group of disease producing agents, which seemed to differ from other forms of living matter in their ability to pass through earthenware filters having a pore diameter smaller than the smallest bacteria then known.

3a. Differentiate between Viruses Pathogenicity and Infectivity

Answer

Viruses pathogenicity is the ability of a virus to cause disease in its multiplicity. While Virus infectivity is the ability of a virus agent to get into a host and multiply.

b. State the type of Genome of Lambda, P22, OX1 74, O6 phages

Answer.

Lambda - Double stranded linear DNA

P22 - Linear, Double stranded DNA

OX1 74 - [+] sense circular single-stranded DNA O6 -

c. Which of the phages in section 3b above are template phages?

d. State the 3 major system for cultivating animal viruses in Virology Laboratory

Answer

* Cell culture

* Embryolated chicken eggs

* Laboratory animals which includes Mice and Monkey

e. With one example each, use diagrams to describe the three possible ways animal viruses enter their cellular host.

4a. State seven (7) materials or equipment used in Virology Laboratory.

Answer

* Safety cabinet

* Centrifuge

* Pipette

* Incubator

- * Egg puncher

- * ELISA plate

- * ELISA washer and reader

b. Describe with a sketch, the replication cycle of Baltimore's class I viruses

Answer

Class II viruses; Single stranded viruses I, examples; B-19 Parvovirus, Canine Parvovirus. The genome replicate in the nucleus of the host / nucleus of infected cells. The S.S DNA serves as template for synthesis of D.S DNA which subsequently serve as template for genome replication.

c. State 5 indicators of replication of animal viruses in cell culture

Answer

- * Cell rounding

- * Giant cell formation

- * Membrane blebbing

- * Syncytium or syncytian formation

- * Granular formation

d. Give two (2) reasons for studying bacteria viruses

Answer

- * To know about their structure.

- * To know about the disease caused.

c. What is an assay, state three (3) viral assays you know.

Answer

An assay is a laboratory procedure to find out the quantity or the quality (presence or absence) of HIV in the blood.

Three viral assays

- * Infectivity assay
- * Plaque assay
- * ELISA - Enzyme Linked Immune Sorbet Assay

5a. Differentiate between viroids, virusoids and prions

Answer

Viroids are molecules of RNA that are not classified as viruses because they lack protein coat. While Virusoids are made up of RNA only. While Prions are infectious protein molecules that do not contain DNA or RNA.

b. State five (5) animal viruses that are vertically spread

c. Besides vertically transmitted, State five (5) other animal viruses and describe mode of transmission of each.

Answer.

- * Dengue Virus - Mosquito bite
- * Ebola viruses - body contact
- * Epstein - Barr Virus - Kissing
- * Polio virus - Spoilt food or water
- * HIV - Sexually transmitted disease

* Influenza virus, Corona virus - Inhalation.

d. Write short notes on Viral capsid, Viral envelope and viral genome.

Answer

* Viral capsid: This serves as a membrane used to protect the genome or genetic material.

* Viral envelope: Some species of viruses envelope themselves in a modified form if one of the cell membrane surrounding an infected host cell, or internal membrane such as nuclear membrane or endoplasmic reticulum, thus, gaining an outer lipid bilayer known as viral envelope.

* Viral genome: This is associated with proteins within a central structure known as a nucleoid. The nucleoid is surrounded by a membrane and two lateral bodies of known functions.

The virus has an outer envelope with a thick layer of protein studded over the surface.

2019/20

2. What is assay in Virology? Mention any two assays in Virology lab.

Answer

Assay in Virology is a laboratory procedure to find out the quantity of HIV in the blood or to find out the presence or absence of HIV in human blood ASSAYS IN VIROLOGY LAB

* PLAQUE ASSAY

* INFECTIVITY ASSAY

3. Differentiate between a plant virus and a viroid.

Answer

Plant virus invade, infect or multiply in higher plants. While Viroids are RNA only and usually infects plants

4. Mention two (2) viral diseases and state the genome type

Answer

* Adeno virus – DNA

* Parvo virus – DNA

5. What two (2) viruses do you know that infects human guts?

6. What is the difference between prophage and provirus?

Answer

Prophage refers to the genome of the virus, integrated into the genome of eukaryotic host cell.

7. What 2 conditions can make a temperate phage lysogenise in its host?

Answer

* Starvation of host

* High MOI (Multiplicity of infection)

* Low incubation temperature.

* Stable clear II protein

8. Draw and label a named enveloped virus

9. Mention three (3) system for propagating animal Virus in Virology lab.

Answer

* Embryolated egg

* Cell culture

* Laboratory animals

10. Differentiate between virus infectivity and pathogenicity.

Answer

Virus infectivity is the ability of a virus to get to the host and multiply. While Virus pathogenicity is the ability of a virus agent to cause disease in its multiplicity.

11. What is maturation steps essential to animal Virus morphogenesis?

12. What is "Central dogma" in biology ?

Answer

DNA → RNA → Protein

13. Where in infected host cell would progeny paramyxovirus and retro assemble?

Answer

They assemble in the cytoplasm.

14. State the contributions of David Baltimore and Max Theiler to virology

Answer

* David Baltimore: independently discovered reversed transcriptase in retro viruses and thus was the first study to refute central dogma in biology.

* Max Theiler: He developed attenuated yellow fever vaccine. He got a Nobel prize award in 1951. The vaccine saved lot of life's and still effective till today.

15. Why is TMV describe as Filterable?

16. Sketch genome replication of Baltimore's class III and VI

17. Mention two (2) plant Viruses having filamentous morphology

Answer

* Potato virus

* Tobacco mosaic virus (TMV).

18. To be infective, a plant virus must have at least * enzyme, * capsid and * Protein genes.

19. Why most viruses be stored freezing cold in Virology lab?

Viruses are stored freezing cold in Virology lab to make them Viable.

20. Differentiate between viral envelope and capsid

21. State two (2) viruses each having linear and circular genomes.

22. What makes different cells readily susceptible to infection by (a) Polio virus (b) TBSV - 2

23. Mention any two (2) peculiarities of viruses.

* They are acellular, i.e., they contain no cytoplasm

* They carry out no metabolism on their own and must replicate using the host cells metabolic machinery.

* They are the smallest infectious agents.

* They contain only one kind of nucleic acid (DNA or RNA) as their genome.

24. Name a virusoid and a viroid

Answer

- * Tomato chlorotic dwarf

- * Apple crinckel

- * Potato spindle fiber viroid

Virusoid

- * Hepatitis delta virus

25. Why would progeny parvoviruses not exit the infected cell by Budding?

26. Mention two (2) viruses each that are air-borne and arthropod borne

Answer.

Air-borne

- * Influenza virus

- * SARS- COV 2

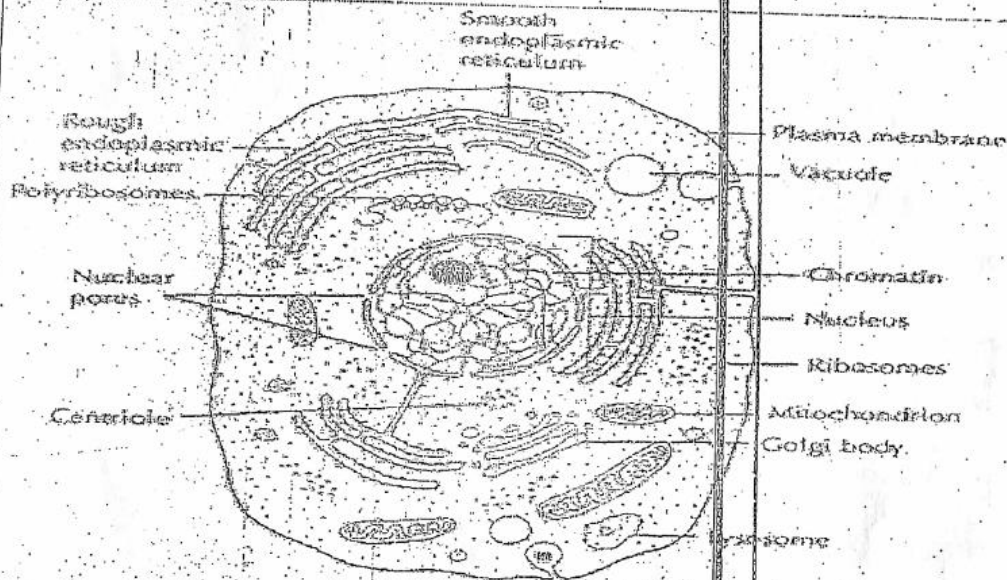
Arthropod- borne

- * Yellow fever virus

PAST TEST QUESTIONS

1. Draw and fully label an animal cell.

Answer



2. In tabular form, name three viruses each of plants, bacteria, and animals.

Answer

Bacteria viruses	Plant Viruses	Animal viruses
Escherichia virus T4	Tobacco mosaic virus (TMV)	Human Immunodeficiency virus (HIV)
Enterobacteria phage T2	Tomato spotted with virus (TSWV)	Ebola virus
Coli phage	Potato virus Y (PSY)	Herpes virus

3. Name 3 scientist and their contributions to the field of Virology.

Answer

In 1970, David Baltimore and Howard Temin independently, discovered reversed transcriptase in retro viruses and that was the first study to refute central dogma in biology. DNA - RNA - Proteins.

In 1917, Felix d'Herelle independently discovered viruses of bacteria and coined the word Bacteriophage. He provided an invaluable means of studying virus replication prior to the advent of cell culture when the only means to study viruses was by infecting whole animals.

In 1898, Frederick Loeffler and Paul Frosch demonstrated FMD (Foot and Mouth disease) which was caused by such filterable agent. They were the first to prove that Viruses infects animals including plants.

4. What 3 conditions makes a template phage exhibit a lysogenic pathway of replication.

Answer

↔ Starvation of host.

↔ Low incubation temperature

↔ High MOI

5. Attempt a classification of plant Viruses based on genom property

Answer

↔ Nature

↔ Segmented or non-segmented

↔ Gene number

6. After nucleic acid injection, what other step does a typical phage undertake during replication cycle.

Answer

↔ DNA replication (Transcription and translation)

↔ Assembly or packaging

↔ Release

7. State four (4) reasons we study viruses.

Answer

↔ To know about their structure

↔ They are good models to study animal cells.

↔ To know how they carry out their life cycle.

↔ To know about their use in research and therapy.

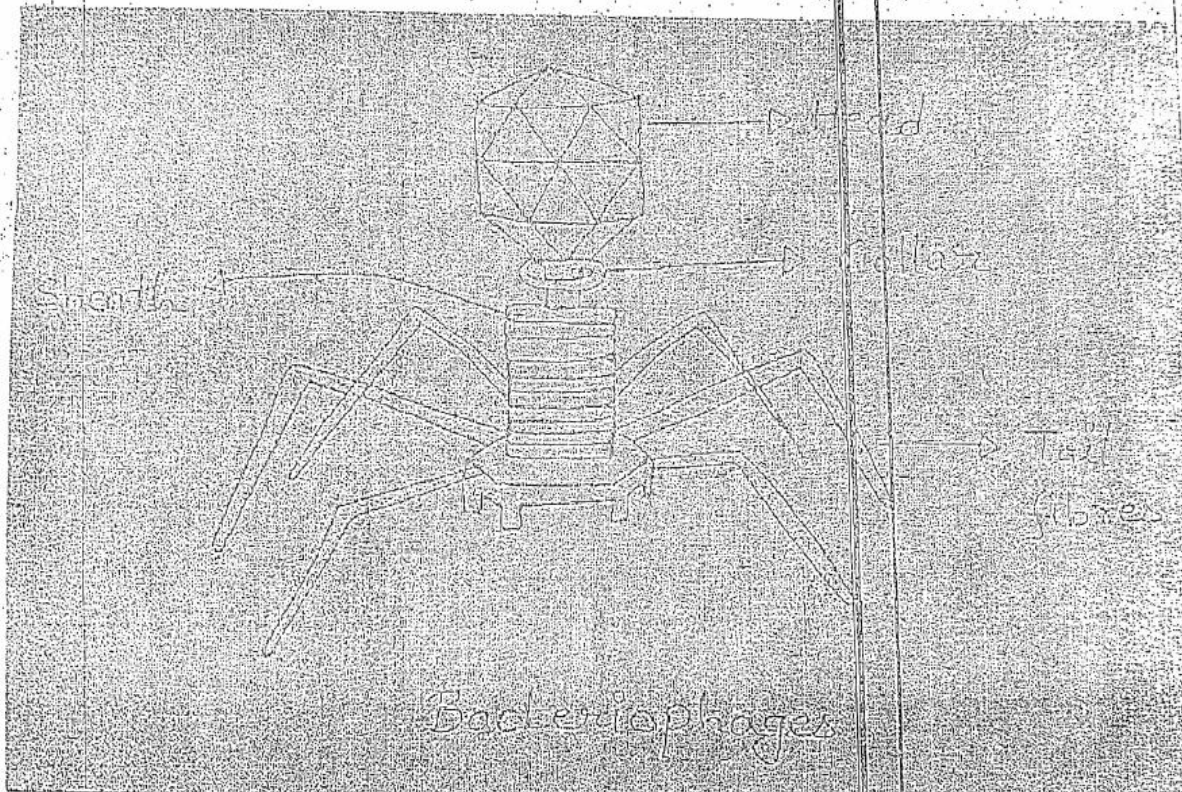
8. Viruses have living and non - attributes. Justify their association.

Answer

Viruses have living attributes because they become living once they gain entry into living host while viruses have non-living attributes because inactive outside the host.

9. Draw a fully labelled typically Bacteria Virus

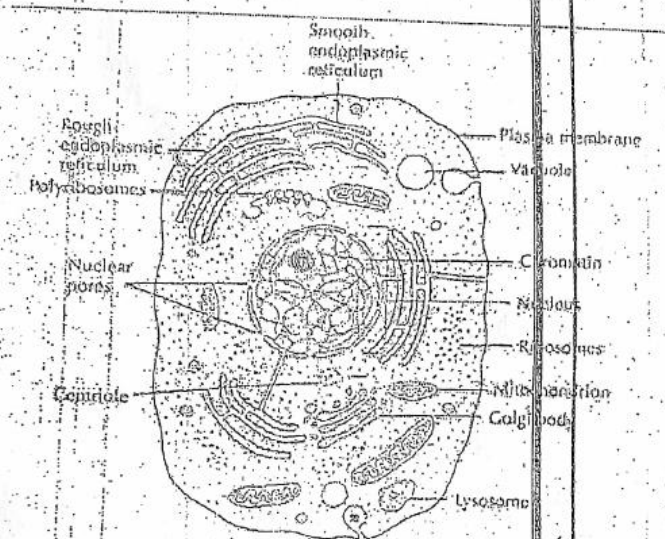
Answer:



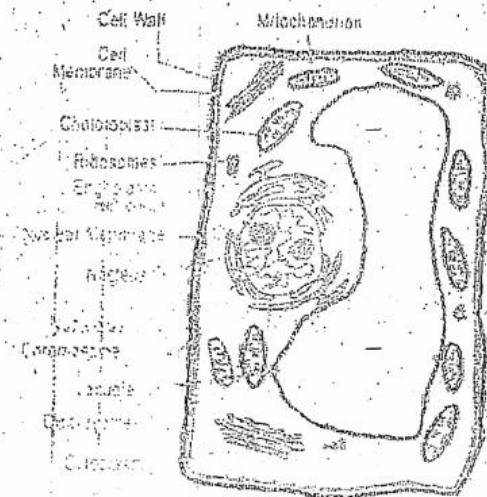
10. Use diagrams to differentiate between plant and animal cells.

Answer

Animal cell



Plant cell



Mcb 203

Past question and answer

2021/2022

Instruction: Answer all question in section A and any two question from section B.

1. Mention three scientist and contribution of each to virology.

ANS: (a) Edward Jenner (1749-1823)----He used cowpox to vaccinate against smallpox.

(b) Louis Pasteur (1822-1895)----He experimented with rabies vaccination and originated the term "virus" means (poison).

(c) Walter Reed: he showed that yellow fever virus is spread by virus. he and his co workers showed that virus could be spread by insect vectors.

2. In a tabular form, state four differences between phages and *rickettsia spp*.

ANS:

3. What is animal virus-induced cytopathic effects (CPEs)?

ANS: This refers structural changes in host cells that are caused by viral invasion.

4. Mention two CPEs:

ANS: (i) Focal Degeneration

(ii) Total Destruction.

5. Mention a named DNA virus and describe the steps for the virus to successfully infect a cell.

ANS: named DNA virus : PARVO VIRUS.

STEPS: (i) Attachment

(ii) Penetrating

(iii) Uncoating

(iv) Genome replication

(v) Assembly

(vi) Maturation

(vii) Release

6. List the orderly steps of T4 phage replication cycle.

ANS: (i) Attachment

(ii) Penetration

(iii) Integration (prophage)

(iv) Replication (binary fission)

7. Name three plant virus species and host of each.

PLANT VIRUS SPECIES-----HOST

(i) Tobacco Mosaic Virus-----Tobacco plant.

(ii) Cucumber Mosaic Virus-----Cucumber Plant.

(iii) Potato Virus Y-----Potato Plant

8. What are the mechanisms by which viruses alter their genotype?

(i) Mutation.

(ii) Recombination.

(iii) Hypermutation.

(iv) Genome segment reassortment.

9. Mention five animal viruses and their transmission route.

ANIMAL VIRUS-----TRANSMISSION ROUTE.

(i) Rabies-----Direct contact.

(ii) Noro virus-----Contact.

(iii) HIV virus-----Sexual Transmitted.

(iv) Psittacosis virus-----Respiration.

(v) Rota viruses-----Fecal-Oral Route.

10. Differentiate between these: viral infectivity; viral pathogenicity; viral disease.

ANS: Viral infectivity is defined as the capacity of viruses to enter the host cell and exploit its resources to replicate and produce progeny infectious viral. while viral pathogenicity pertains to the ability of a pathogenic agent to cause disease. and Viral disease are infections caused by viruses.

11. What is the determinant of poliovirus infectivity?

ANS: the receptor

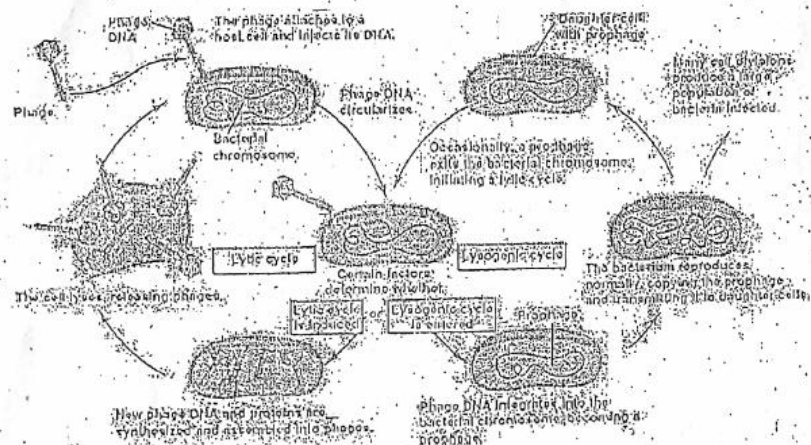
12: What determines the susceptibility of a given animal cell to a given viral infection?

ANS: Presence of a specific receptor for that viral infection.

SECTION B

Answer any two question below;

1. a). Using a detailed diagram, described phage lysogenic pathway.



b). With diagrams, differentiate between the three mechanism of animal virus entry into susceptible cells.

c). List three difference between viruses and E.coli.

i). E.coli are living organism while virus can replicate only within the host cell.

ii). E.coli reproduce asexually by binary fission while virus inject their genome in the host genome and make copies.

iii). E.coli cell is made up of peptidoglycan while bacteria do not contain a cell wall.

2. a). What are viroids ; mention four viroid and host of each. Viroids are infectious entity affecting plants, smaller than a virus and consisting only of nucleic acid without a protein coat.

1 – Pospiviroidae: Members of the pospiviroidae are named after the potato spindle tuber viroid and the host is potato.

2 – Avsunviroidae: is named after avocado sunblotch viroid

3-potato sprint disease: the host is potato.

4- Citrus Exocurtis: the host is citrus.

1. Papovaviruses: nucleic acid hybridization is the clinical sample needed.

2. Simian Virus-40 (SV40): can be identified using primer or PCR.

3. Adenoviruses: can be identified using antigen detection or PCR.

4. Herpesviruses: can be identified using PCR.

c). List some methods of purifying viruses.

i). Chromatography

ii). Flow field- fractional

iii). Dialysis

iv). Filtration.

3. a). Differentiate between primary cell culture, cell strain and cell line.

Primary cell culture is the culture of cells directly isolated from parental tissue of interest; whereas cell line is the culture of cells originated from a primary cell culture, which is generally used to expand cell population and prolong life span. A cell strain is a specific cell obtained from a cell line, and the cell type refers to the finally differentiated cell.

b). Mention any three disease caused by prions.

1. scrapie (sheep),
2. chronic wasting disease (deer, elk, moose)
3. mad cow disease (cattle)

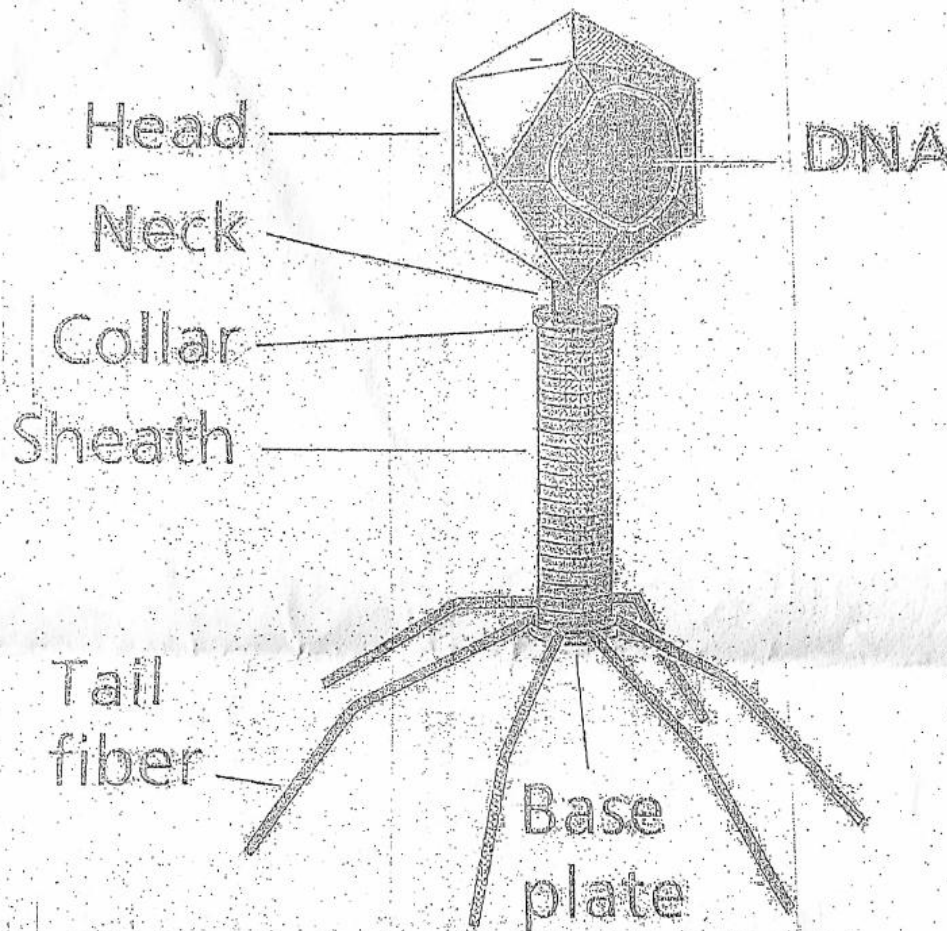
c). List the Baltimore classification of viruses with an example each.

- I: dsDNA viruses (e.g. Adenoviruses, Herpesviruses, Poxviruses)
- II: ssDNA viruses (+)sense DNA (e.g. Parvoviruses)
- III: dsRNA viruses (e.g. Reoviruses)
- IV: (+)ssRNA viruses (+)sense RNA (e.g. Picornaviruses, Togaviruses)
- V: (-)ssRNA viruses (-)sense RNA (e.g. Orthomyxoviruses, Rhabdoviruses)
- VI: ssRNA-RT viruses (+)sense RNA with DNA intermediate in life-cycle (e.g. Retroviruses)
- VII: dsDNA-RT viruses (e.g. Hepadnaviruses)

4. a). Mention three host systems for culturing viruses.

- i). Bacteria.
- ii). Archaea
- iii). Eukarya

b). Draw a well T4 phage.



c). In a tabular form, list 3 difference between viruses and prion.

1. virus is composed of both nucleic acids and proteins while prion is composed only of a protein. 2. Prion lacks DNA or RNA while viruses possess nucleic acids either DNA or RNA. 3. Virus has capsid but prion lack capsid.